

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C02: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:30

Annexure A

Dry Run Evaluation

Code of Conduct:

I Priyanka G/192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

20+20+24+20+14

98

AP

Annexure A
Dry Run Evaluation

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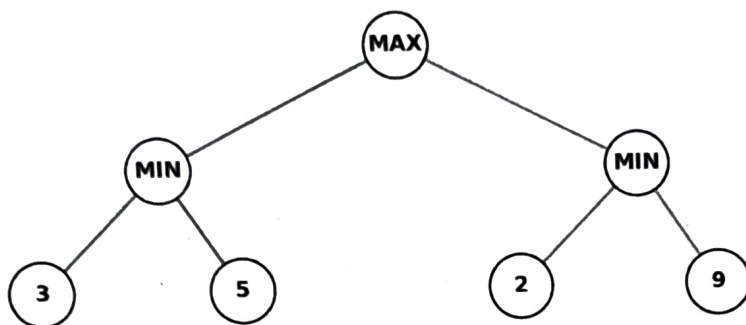
1. Question 16 - Cryptarithmic CSP

SEND + MORE = MONEY. Each letter represents a unique digit from 0-9 and M cannot be 0. Perform constraint propagation, find a valid assignment and verify the addition.

2. Question 17 - Minimax Dry Run

Perform the Minimax algorithm on the displayed game tree.

Calculate the value of each MIN node, the MAX root value and the move selected by MAX.



Node	Children Values	Value
1		
2		
3		
4		
5		

3. Question 18 - Minimax with Deeper Tree

Consider a MAX root with two MIN children. Each MIN child has two MAX children. The leaf values from left to right are: 3,5 | 2,9 | 4,6 | 1,7.

Perform a complete Minimax dry run and identify the optimal move.

Node	Children Values	Value
1		
2		
3		
4		
5		
6		
7		

1) CRYPTARITHMETIC CSP.Given: SEND + MORE = MONEYEach letter represents a unique digit, and $M \neq 0$.

Align the addition:

$$\begin{array}{r} \text{SEND} \\ + \text{MORE} \\ \hline \text{MONEY} \end{array}$$

Introduce carry variables C_1, C_2, C_3, C_4 .

From the right to left:

Column 1: $D + E = Y + 10C_1$

Column 2: $N + R + C_1 = E + 10C_2$

Column 3: $E + 0 + C_2 = N + 10C_3$

Column 4: $S + M + C_3 = 0 + 10C_4$

Since, the result has five digits, the leftmost carry must be:

$$C_4 = M = 1, \therefore M = 1.$$

Constraint propagation gives the standard valid assignment:

$$S=9, E=5, N=6, D=7, M=1, O=0, R=8, Y=2.$$

So:

$$\begin{array}{r} 9 \ 5 \ 6 \ 7 \\ + 1 \ 0 \ 8 \ 5 \\ \hline 1 \ 0 \ 6 \ 5 \ 2 \end{array}$$

Verification:

$$9567 + 1085 = 10652$$

Final Assignment:

Letter	S	E	N	D	M	O	R	Y
Digit	9	5	6	7	1	0	8	2

All letters have unique digits and $M=1 \neq 0$, so the solution is valid

In the cryptarithmic problem, each letter is assigned a unique digit from 0-9, and the constraints are propagated from the units column to the leftmost column while ensuring that M is not zero.

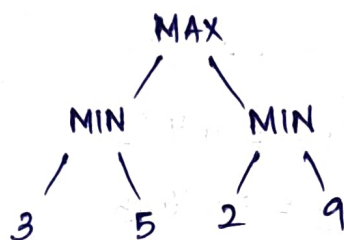
2) MINIMAX DRY RUN.

The Minimax algorithm evaluates the game tree from the leaf nodes upwards. MIN nodes select the minimum value among their children, while the MAX root selects the maximum value among its children.

Minimax Rule:

$$\text{MIN} = \min(\text{child values}), \text{MAX} = \max(\text{child values})$$

Given Tree:



Step 1: Children are 3 and 5 - left MIN Node

$$\text{MIN}(3, 5) = 3$$

∴ The left MIN node = 3

Step 2: Right MIN Node - Children are 2 and 9.

$$\text{MIN}(2, 9) = 2$$

∴ The right MIN node = 2

Step 3: MAX Root - The MAX root receives the values 3 and 2.

$$\text{MAX}(3, 2) = 3$$

∴ MAX Root value = 3.

Hence, the left branch gives 3 and the right branch gives 2, Max selects the left branch, Move selected by MAX = left MIN node.

$\therefore$, the Minimax value of the root is 6.
 Hence, the right branch has higher value, MAX selects the right MIN subtree.

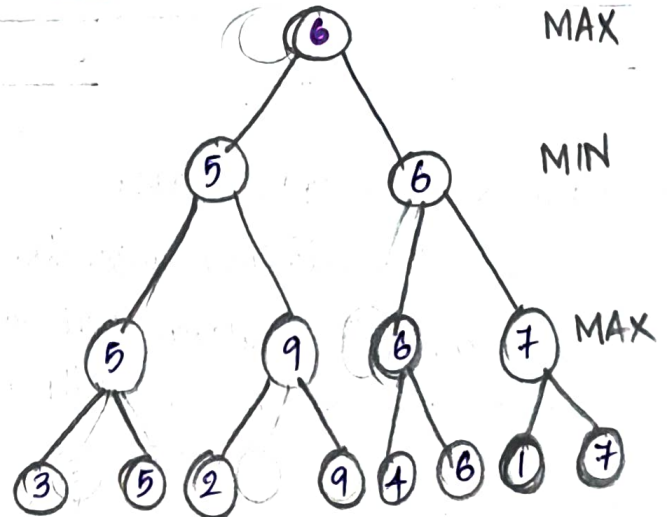
Optimal Move = Right MIN node

Minimax value = 6.

Complete Dry-Run Table

Node	Children Values	Value
1	5, 9	5
2	6, 7	6
3	3, 5	5
4	2, 9	9
5	4, 6	6
6	1, 7	7
7	5, 6	6

Final tree:



Hence, max chooses the right subtree because its minimax value is 6, which is greater than left subtree's value of 5.